
SLEEP CYCLE PRODUCTIVITY DATASET ANALYSIS



Analyzing Sleep Patterns and Their Relationship
with Productivity



A data-driven analysis of how sleep duration, sleep quality,
lifestyle factors and habits influence productivity.

BY

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INTRODUCTION

Sleep plays an important role in maintaining daily productivity, mood, and overall well-being. This project focuses on analyzing sleep patterns and their relationship with productivity using a dataset containing information about sleep habits, lifestyle factors, and productivity levels.

The analysis explores various factors that may influence productivity, including sleep duration, sleep quality, stress levels, exercise, caffeine consumption, screen time, and work-related habits. Through data cleaning, exploratory data analysis (EDA), and data visualization using Power BI, the project aims to identify meaningful patterns and relationships within the dataset. The analysis provides useful insights into how sleep and lifestyle habits are associated with productivity and helps in understanding the factors that may contribute to improved daily performance and well-being.

AIM

The main aim of this project is to explore the relationship between sleep habits, lifestyle factors, and daily productivity using data analysis and visualization techniques. The analysis focuses on understanding how different aspects of sleep and daily routines are connected with productivity and overall well-being. Specifically, the analysis aims to:

1. **Understand Sleep Patterns:** Analyse sleep duration, sleep quality, and other sleep-related factors to identify common patterns.
2. **Study Productivity:** Explore productivity levels and understand how they vary with different sleep and lifestyle habits.
3. **Analyse Lifestyle Factors:** Examine the influence of stress, exercise, caffeine intake, screen time, and work-related habits on sleep and productivity.
4. **Identify Relationships and Trends:** Find meaningful patterns and relationships between sleep, lifestyle choices, and productivity using data analysis and visualizations.
5. **Generate Useful Insights:** Use the findings to better understand how healthy sleep habits and daily routines may contribute to improved productivity and well-being.

Overall, this project aims to turn the available data into simple and meaningful insights that can help in understanding the importance of sleep and lifestyle habits in maintaining better daily productivity.

OBJECTIVES

1. Data Collection:

Collected and organized sleep and productivity data containing information such as age, gender, sleep duration, sleep quality, exercise, caffeine intake, screen time, work hours, productivity, mood, and stress levels. The data was prepared in a structured format for further analysis and visualization.

2. Data Cleaning and Preprocessing:

Performed data cleaning and preprocessing using Python to improve the quality and reliability of the dataset. This included handling missing values, removing duplicate records, checking data types, identifying inconsistencies, and preparing the data for analysis.

3. Python-Based Data Analysis:

Used Python libraries such as Pandas and NumPy to analyze the dataset and extract meaningful insights. Applied functions and operations such as `describe()` , `value_counts()`, `groupby()` , `loc[]` , and `iloc[]` to examine the characteristics and patterns within the data.

4. Exploratory Data Analysis (EDA):

Conducted exploratory data analysis using Python to understand patterns and relationships between sleep habits and productivity. The analysis focused on sleep

duration, sleep quality, productivity score, mood, stress levels, exercise, caffeine consumption, screen time, and work hours.

5. Data Visualization and Dashboard Development:

Developed an interactive Power BI dashboard to visualize important sleep and productivity metrics. Created charts, graphs, KPI cards, and interactive filters to present insights related to sleep patterns, productivity, stress, mood, caffeine intake, exercise, and screen exposure.

6. Insight Generation:

Generated meaningful analytical insights from the dataset using Python-based analysis and Power BI visualizations. The findings help identify factors associated with better sleep, improved productivity, and healthier daily routines.

DATA OVERVIEW

Column name	Data Types
Date	Datetime64[ns
Person_ID	int64
Age	int64
Gender	Object
Sleep Start Time	float64
Sleep End Time	float64
Total Sleep Hours	float64
Sleep Quality	int64
Exercise (mins/day)	int64
Caffeine Intake (mg)	int64
Screen Time Before Bed (mins)	int64
Work Hours (hrs/day)	float64
Productivity Score	int64
Mood Score	int64
Stress Level	int64
Sleep Status	Object
Caffeine Intake Level	Object
Screen Exposure Level	Object

DATA ANALYSIS

1. Descriptive Analysis:

This section provides an overview of the key characteristics of sleep patterns and productivity levels within the dataset. By examining metrics such as average sleep duration, sleep quality, productivity score, stress levels, screen time, caffeine consumption, exercise, and work hours, we gain a better understanding of the overall lifestyle and productivity patterns of the individuals.

2. Sleep Pattern Analysis:

This section focuses on analyzing different sleep-related factors and their patterns. Sleep duration and sleep quality are examined to identify common sleeping patterns and understand how variations in sleep habits are associated with daily productivity, mood, and overall well-being.

3. Lifestyle and Productivity Analysis:

This section examines the relationship between lifestyle factors and productivity levels. Factors such as exercise, caffeine consumption, screen time, stress levels, and work hours are analyzed to identify patterns that may be associated with higher or lower productivity scores.

4. Relationship Analysis:

This section investigates the relationships between sleep, lifestyle, and productivity variables. By comparing sleep duration and quality with productivity, mood, and stress levels, the analysis aims to identify meaningful associations and patterns that provide a deeper understanding of factors related to individual productivity.

5. Productivity Analysis:

This section analyzes productivity scores across different sleep and lifestyle conditions. The analysis identifies productivity patterns based on sleep quality, sleep duration, exercise, stress, caffeine intake, screen time, and work hours to determine which factors are associated with variations in productivity.

6. Visualization and Dashboard Analysis:

This section presents the key findings through interactive visualizations and dashboards developed using Power BI. KPI cards, charts, graphs, and filters are used to provide a clear and interactive view of sleep patterns, productivity levels, lifestyle factors, and their relationships.

QUESTIONS

1. Total number of unique individuals:

```
df['Person_ID'].nunique()
```

2. Frequency of each individual:

```
df['Person_ID'].value_counts()
```

3. Unique Person_ID count:

```
df['Person_ID'].value_counts()[df['Person_ID'].value_counts() == 1].sum()
```

4. Person_ID count appearing twice:

```
print(df['Person_ID'].value_counts()[df['Person_ID'].value_counts() == 2])
```

```
print((df['Person_ID'].value_counts() == 2).sum())
```

5. Gender descriptive statistics:

```
df['Gender'].describe()
```

6. Number of individuals by Gender

```
df['Gender'].value_counts()
```

7. Mean of numerical columns:

```
print("Mean:")
```

```
df.select_dtypes(include='number').mean()
```

8. Median of numerical columns:

```
print("Median")
```

```
df.select_dtypes(include="number").median()
```

9. Sleep Start Time statistics:

```
df['Sleep Start Time'].describe()
```

10. Standard deviation of numerical columns:

```
print("Std:")
```

```
df.select_dtypes(include="number").std()
```

11. Mean Productivity Score Grouped by Sleep Quality

```
df.groupby('Sleep Quality')['Productivity  
Score'].mean()
```

12. Extracting All Records from the Stress Level and Productivity Score Columns

```
df.loc[:,['Stress Level','Productivity Score']]
```

13. Statistical Summary of Caffeine Intake:

```
df['Caffeine Intake (mg)'].describe()
```

14. Extracting the First 10 Records from the 3rd and 7th Columns:

```
df.iloc[:10,[2,6]]
```

15. Analysis of Sleep Quality for Screen Time Before Bed (mins) Greater Than or Equal to 100

```
df.loc[df["Screen Time Before Bed (mins)"]
```

```
>=100,['Screen Time Before Bed (mins)','Sleep
```

```
Quality']]
```

16. Mean Productivity Score Grouped by Stress Level

```
df.groupby('Stress Level')['Productivity
```

```
Score'].mean()
```

17. Analysis of Productivity Score Greater Than 8 for Total Sleep Hours Greater Than 6

```
df.loc[ (df['Total Sleep Hours'] >6) &
```

```
(df['Productivity Score'] >8),
```

```
['Total Sleep Hours','Productivity Score']
```

```
]
```

18. Statistical Summary and Distribution Analysis of the Productivity Score

```
df['Productivity Score'].describe()
```

19. Summary Statistics of Productivity Score Grouped by Sleep Quality

```
df.groupby('Sleep Quality')['Productivity  
Score'].agg(['mean','median','min','max'])
```

CONCLUSION

The analysis reveals important insights into sleep patterns, lifestyle habits, and productivity levels. The dataset was analyzed to understand the relationship between sleep duration, sleep quality, caffeine intake, screen time before bed, exercise, stress, mood, and productivity. Different sleep-status groups showed variations in average productivity, while caffeine intake and screen-exposure categories also showed differences. The correlation analysis indicates that the individual numerical factors have only weak linear relationships with productivity. Overall, the analysis suggests that productivity is influenced by multiple lifestyle and behavioral factors rather than a single variable. These insights can help individuals and organizations better understand sleep and lifestyle patterns and support data-driven approaches to improving productivity and well-being.

1. Total number of unique individuals:

The dataset contains 3,858 unique individuals based on Person_ID. This provides the total number of distinct individuals included in the sleep and productivity analysis.

2. Frequency of Each Individual:

The Person_ID frequency analysis shows 3,858 unique individuals, with each individual appearing between 1 and 4 times in the dataset. This indicates that some individuals have multiple records, while others appear only once.

Person_ID	
9989	4
7200	4
5887	4
5597	4
7073	4
..	
1594	1
5192	1
7134	1
4205	1
2304	1

Name: count, Length: 3858, dtype: int64

3. Unique Person_ID Count:

The analysis shows that 2,904 individuals appear only once in the dataset. This indicates that these individuals have a single record in the available data.

4. Person_ID Count Appearing Twice:

The analysis shows that 782 individuals appear exactly twice in the dataset. This indicates that these individuals have two records in the available data.

Person_ID
4843 2
8526 2
2853 2
5389 2
3255 2
..
7612 2
5122 2
1617 2
8191 2
6262 2

Name: count, Length: 782, dtype: int64

782

5. Gender descriptive statistics:

The dataset contains 5,000 records across 3 gender categories, with Male being the most frequent category at 1,718 records. This shows that the dataset has a diverse representation of genders.

count	5000
unique	3
top	Male
freq	1718

Name: Gender, dtype: object

6. Number of individuals by Gender

The dataset contains 1,718 Male, 1,675 Female, and 1,607 Other individuals. Male is the most represented gender, while Other has the lowest count.

Gender
Male 1718
Female 1675
Other 1607

Name: count, dtype: int64

7. Mean of Numerical Columns:

The average age is 38.59 years, with an average sleep duration of 6.97 hours and sleep quality of 5.52. The average productivity score is 5.64, while average caffeine intake, screen time, and exercise are 146.71 mg, 91.42 minutes, and 43.96 minutes respectively.

Person_ID	5527.152200
Age	38.590600
Sleep Start Time	22.005350
Sleep End Time	4.980262
Total Sleep Hours	6.974902
Sleep Quality	5.520800
Exercise (mins/day)	43.962600
Caffeine Intake (mg)	146.711400
Screen Time Before Bed (mins)	91.421200
Work Hours (hrs/day)	7.987873
Productivity Score	5.644200
Mood Score	5.370600
Stress Level	5.548000

dtype: float64

8. Median of Numerical Columns:

The median age is 39 years, with median sleep duration of 6.96 hours and sleep quality of 5. The median productivity score is 6, while median caffeine intake, screen time, and exercise are 144 mg, 92 minutes, and 44 minutes respectively.

Person_ID	5603.000000
Age	39.000000
Sleep Start Time	22.020000
Sleep End Time	4.970000
Total Sleep Hours	6.960000
Sleep Quality	5.000000
Exercise (mins/day)	44.000000
Caffeine Intake (mg)	144.000000
Screen Time Before Bed (mins)	92.000000
Work Hours (hrs/day)	7.998295
Productivity Score	6.000000
Mood Score	5.000000
Stress Level	6.000000

dtype: float64

9. Sleep Start Time statistics:

The average sleep start time is approximately 22:00 (10:00 PM), with most individuals starting sleep between 21:02 and 23:00. The sleep start time ranges from 20:00 to 23:59, showing that most individuals go to bed during the late evening.

count	5000.000000
mean	22.005350
std	1.157591
min	20.000000
25%	21.020000
50%	22.020000
75%	23.000000
max	23.980000

Name: Sleep Start Time, dtype: float64

10. Standard Deviation of Numerical Columns:

The highest variation is observed in Person_ID, caffeine intake (85.83 mg), and screen time before bed (52.08 minutes). Productivity score has a standard deviation of 2.87, indicating moderate variation in productivity levels among individuals.

Person_ID	2595.391771
Age	12.302799
Sleep Start Time	1.157591
Sleep End Time	1.862508
Total Sleep Hours	1.454033
Sleep Quality	2.863845
Exercise (mins/day)	25.798541
Caffeine Intake (mg)	85.831390
Screen Time Before Bed (mins)	52.079123
Work Hours (hrs/day)	2.275734
Productivity Score	2.872221
Mood Score	2.835783

Stress Level	2.888419
--------------	----------

dtype: float64

11. Mean Productivity Score Grouped by Sleep Quality:

The average productivity score varies across sleep-quality levels, ranging from 5.32 to 5.81. The highest average productivity is observed at sleep quality 5 (5.81), while the lowest is at sleep quality 4 (5.32).

Sleep Quality

1	5.647917
2	5.731610
3	5.655102
4	5.318898
5	5.806142
6	5.785276
7	5.653772
8	5.651731
9	5.464583
10	5.717850

Name: Productivity Score, dtype: float64

12. Extracting All Records from the Stress Level and Productivity Score Columns

The loc function extracts all 5,000 records from the Stress Level and Productivity Score columns. This

Stress Level	Productivity Score	
0	6	8
1	7	10
2	10	10
3	2	8
4	9	8
...	...	...
4995	1	5
4996	2	5
4997	1	2
4998	6	8
4999	4	6

allows the relationship between stress levels and productivity scores to be examined.

5000 rows × 2 columns

13. Statistical Summary of Caffeine Intake:

The average caffeine intake is 146.71 mg, with a median of 144 mg and a standard deviation of 85.83 mg.

Caffeine intake ranges from 0 to 299 mg, showing considerable variation among individuals.

count	5000.00000
mean	146.71140
std	85.83139
min	0.00000
25%	73.00000
50%	144.00000
75%	220.00000
max	299.00000

Name: Caffeine Intake (mg), dtype: float64

14. Extracting the First 10 Records from the 3rd and 7th Columns:

The `iloc` function extracts the first 10 records from the 3rd and 7th columns, representing Age and Total Sleep Hours. This provides a quick view of age and sleep duration for the first 10 individuals.

Age	Total Hours	Sleep
0	32	5.28
1	41	5.41
2	20	5.35

Age	Total Sleep Hours	
3	37	7.55
4	46	6.75
5	38	8.64
6	18	8.03
7	26	6.35
8	31	7.31
9	49	7.52

15. Analysis of Sleep Quality for Screen Time Before Bed (mins) Greater Than or Equal to 100

The loc function filters the records where Screen Time Before Bed (mins) is greater than or equal to 100. It then extracts the Screen Time Before Bed (mins) and Sleep Quality columns. This provides a focused view of individuals who spend at least 100 minutes on screens before bed and helps examine their corresponding sleep quality.

Screen Time Before Bed (mins)	Sleep Quality	
0	116	3

Screen Time Before Bed (mins)	Sleep Quality	
5	123	10
6	122	3
7	171	8
10	128	7
...	...	...
4992	142	3
4994	100	5
4996	177	4
4997	139	6
4998	161	9

2311 rows × 2 columns

16. Mean Productivity Score Grouped by Stress Level

The groupby function groups the records according to Stress Level and calculates the mean Productivity Score for each stress level. This helps compare the average productivity scores across different levels of stress.

Stress Level	
1	5.594000
2	5.914894

3	5.543353
4	5.763052
5	5.259414
6	5.556000
7	5.838776
8	5.614141
9	5.841210
10	5.518234

Name: Productivity Score, dtype: float64

17. Analysis of Productivity Score Greater Than 8 for Total Sleep Hours Greater Than 6

The loc function filters the records where Total Sleep Hours is greater than 6 and Productivity Score is greater than 8. It then displays the Total Sleep Hours and Productivity Score for the matching individuals.

Total Sleep Hours	Productivity Score	
10	6.12	9
15	7.95	9
17	7.71	9
22	7.58	10
23	6.60	9

Total Sleep Hours	Productivity Score	
...	...	...
4977	6.16	10
4983	9.30	9
4985	7.22	10
4987	6.14	10
4990	8.59	10

767 rows × 2 columns

The output shows values such as 6.12 hours of sleep with a productivity score of 9, 7.95 hours with a score of 9, 7.71 hours with a score of 9, 7.58 hours with a score of 10, and 6.60 hours with a score of 9. Overall, 767 records satisfy both conditions, showing individuals who sleep for more than 6 hours and have a productivity score above 8.

18. Statistical Summary and Distribution Analysis of the Productivity Score

The describe() function gives a summary of the Productivity Score for 5,000 records. The average score is 5.64, with scores ranging from 1 to 10. The

median score is 6. The 25th percentile is 3, and the 75th percentile is 8, showing that most scores fall between 3 and 8. The standard deviation is 2.87, indicating some variation in productivity scores.

count	5000.000000
mean	5.644200
std	2.872221
min	1.000000
25%	3.000000
50%	6.000000
75%	8.000000
max	10.000000

Name: Productivity Score, dtype: float64

**19.Summary Statistics of Productivity Score
Grouped by Sleep Quality**

mean	median	min	max	
Sleep Quality				
1	5.647917	6.0	1	10
2	5.731610	6.0	1	10
3	5.655102	6.0	1	10
4	5.318898	5.0	1	10

	mean	median	min	max	
Sleep Quality					
5	5.806142	6.0	1	10	
6	5.785276	6.0	1	10	
7	5.653772	6.0	1	10	
8	5.651731	6.0	1	10	
9	5.464583	5.0	1	10	
10	5.717850	6.0	1	10	

The `groupby()` function groups the data based on Sleep Quality and calculates the mean, median, minimum, and maximum Productivity Score for each group. The mean productivity scores range from 5.32 to 5.81. The median score is mostly 6, while the minimum and maximum scores are 1 and 10 for all sleep quality levels. This shows how productivity scores vary across different sleep quality levels.